Joins

```
SQL*Plus: Release 11.2.0.1.0 Production on Tue Aug 11 02:31:35 2026

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Enter user-name: A020@db2021
Enter password:

Connected to:
Oracle Database 11g Release 11.2.0.1.0 - Production
SQL> create table department
  2  (
  3  dept_id number(3) primary key,
  4  dept_name varchar2(30)
  5  );

Table created.

SQL> insert into department values(101,'cse');

1 row created.

SQL> insert into department values(102,'ece');

1 row created.

SQL> insert into department values(103,'it');

1 row created.

SQL> create table student6(
  2  student_id number(5) primary key,
  3  student_name varchar2(30),
  4  marks number(3),
  5  dept_id number(3),
  6  constraint fk_dept foreign key (dept_id) references department(dept_id)
  7  );

Table created.

SQL> insert into student6 values (11,'nisha',85,101);

1 row created.

SQL> insert into student6 values(12,'yash',90,102);

1 row created.

SQL> insert into student6 values(13,'kush',85,103);

1 row created.

SQL> insert into student6 values(14,'jay',89,101);

1 row created.

SQL> select s.student_id,s.student_name,d.dept_name from student6 s inner join department d on s.dept_id=d
    .dept_id;

STUDENT_ID STUDENT_NAME                   DEPT_NAME
---------- ------------------------------ ------------------------------
        11 nisha                          cse
        12 yash                           ece
        13 kush                           it
        14 jay                            cse

SQL> select s.student_name,d.dept_name from student6 s left outer join department d on s.dept_id=d.dept_id
    ;

STUDENT_NAME                   DEPT_NAME
------------------------------ ------------------------------
jay                            cse
nisha                          cse
yash                           ece
kush                           it
SQL> select s.student_name,d.dept_name from student6 s right outer jopin department d on s.dept_id=d.dept_
```

```
    id;
select s.student_name,d.dept_name from student6 s right outer jopin department d on s.dept_id=d.dept_id
                                                              *
ERROR at line 1:
ORA-00905: missing keyword


SQL> select s.student_name,d.dept_name from student6 s right outer join department d on s.dept_id=d.dept_i
    d;

STUDENT_NAME                   DEPT_NAME
------------------------------ ------------------------------
nisha                          cse
yash                           ece
kush                           it
jay                            cse

SQL> select s.student_name,d.dept_name from student6 s full outer join department d on s.dept_id=d.dept_id
    ;

STUDENT_NAME                   DEPT_NAME
------------------------------ ------------------------------
nisha                          cse
yash                           ece
kush                           it
jay                            cse

SQL> CREATE TABLE EMPLOYEE6
  2  (
  3   EMP_ID NUMBER(5) PRIMARY KEY,
  4   EMP_NAME VARCHAR2(30),
  5   MANAGER_ID NUMBER(5)
  6  );

Table created.

SQL> INSERT INTO EMPLOYEE VALUES (1,'JAY',NULL);
INSERT INTO EMPLOYEE VALUES (1,'JAY',NULL)
            *
ERROR at line 1:
ORA-00947: not enough values


SQL> INSERT INTO EMPLOYEE VALUES (2,'GOPI',1);
INSERT INTO EMPLOYEE VALUES (2,'GOPI',1)
            *
ERROR at line 1:
ORA-00947: not enough values


SQL> INSERT INTO EMPLOYEE VALUES (3,'HARI',1);
INSERT INTO EMPLOYEE VALUES (3,'HARI',1)
            *
ERROR at line 1:
ORA-00947: not enough values


SQL> CREATE TABLE EMPLOYEE6
  2  (
  3   EMP_ID NUMBER(5) PRIMARY KEY,
  4   EMP_NAME VARCHAR2(30),
  5   MANAGER_ID NUMBER(5)
  6  );
CREATE TABLE EMPLOYEE6
             *
ERROR at line 1:
ORA-00955: name is already used by an existing object


SQL> INSERT INTO EMPLOYEE6 VALUES (1,'JAY',NULL);

1 row created.

SQL> INSERT INTO EMPLOYEE6 VALUES (2,'GOPI',1);

1 row created.
SQL> INSERT INTO EMPLOYEE6 VALUES (3,'HARI',1);
```

```
1 row created.

SQL> SELECT E.EMP_NAME EMPLOYEE, M.EMP_NAME MANAGER
  2  FROM EMPLOYEE6 E
  3  LEFT JOIN EMPLOYEE M
  4   ON E.MANAGER_ID = M.EMP_ID;
 ON E.MANAGER_ID = M.EMP_ID
                   *
ERROR at line 4:
ORA-00904: "M"."EMP_ID": invalid identifier


SQL> SELECT STUDENT_NAME
  2  FROM STUDENT6
  3  WHERE DEPT_ID IN
  4  (
  5   SELECT DEPT_ID
  6   FROM DEPARTMENT
  7   WHERE DEPT_NAME = 'CSE'
  8  );

no rows selected

SQL> SELECT STUDENT_NAME, MARKS, DEPT_ID
  2  FROM STUDENT6 S1
  3  WHERE MARKS =
  4  (
  5   SELECT MAX(MARKS)
  6   FROM STUDENT6 S2
  7   WHERE S1.DEPT_ID = S2.DEPT_ID
  8  );

STUDENT_NAME                        MARKS    DEPT_ID
------------------------------ ---------- ----------
yash                                   90        102
kush                                   85        103
jay                                    89        101

SQL>
```